GE Drive Systems

INSTRUCTIONS

531X171TMAA_G2
4TB TERMINAL BOARD CARD

7V11CTB Option Kit

IMPORTANT
INFORMATION

CAUTION: To ensure proper operation of the QIII DC Controller, these instructions must be followed for proper installation and set-up of this Option Kit.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

SAFETY PRECAUTIONS

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

NOTE: *Read all WARNINGS, CAUTIONS and NOTES in the QIII DC Controller Instruction Book (GEK-45167) prior to installing this Option Kit.*

NOTE: *Read and heed all WARNING, CAUTION, and NOTE labels posted on the equipment prior to installing this Option Kit.*

KIT DESCRIPTION AND CONTENTS

DESCRIPTION:

This Option Kit provides convenient termination points for external interface to the controller. Connections to the optional Application card (Kit 7V11CAC) are made through 4TB. In addition, three 115 VAC relays are provided on this terminal board card.

CONTENTS:

The kit contains the following components (in addition to these instructions):

Component	Part Number	Qty.
4TB Terminal Board Card	531X171TMAA_G2	1
Mounting Standoffs	----	4
Mounting Hardware	----	--
Insulation Barrier	36A358215AA002	1

NOTE: *If Balun Option Kit (7V11CBA) is also to be installed on the controller, It should be placed onto the appropriate mounting screws* ***prior*** *to installing this 4TB Terminal Board Option Kit.*

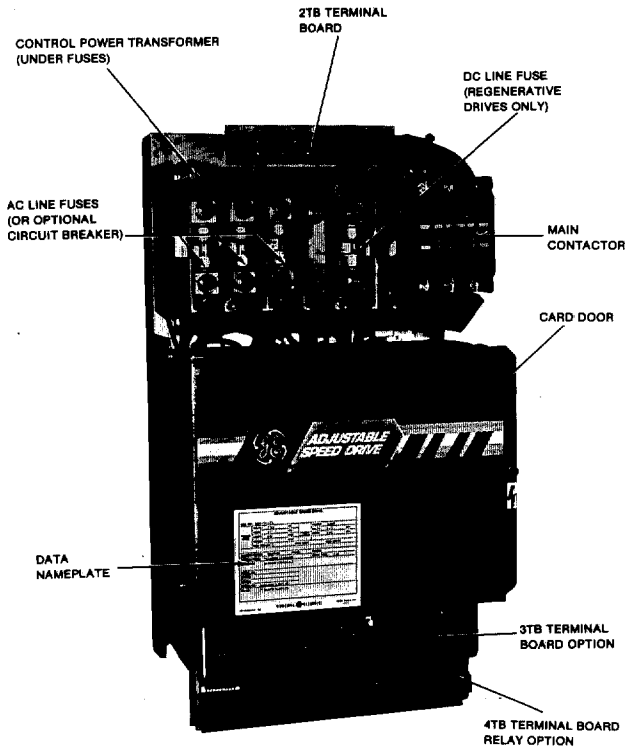


Figure 1. 4TB MOUNTING LOCATION ("C" and "G" Frames)

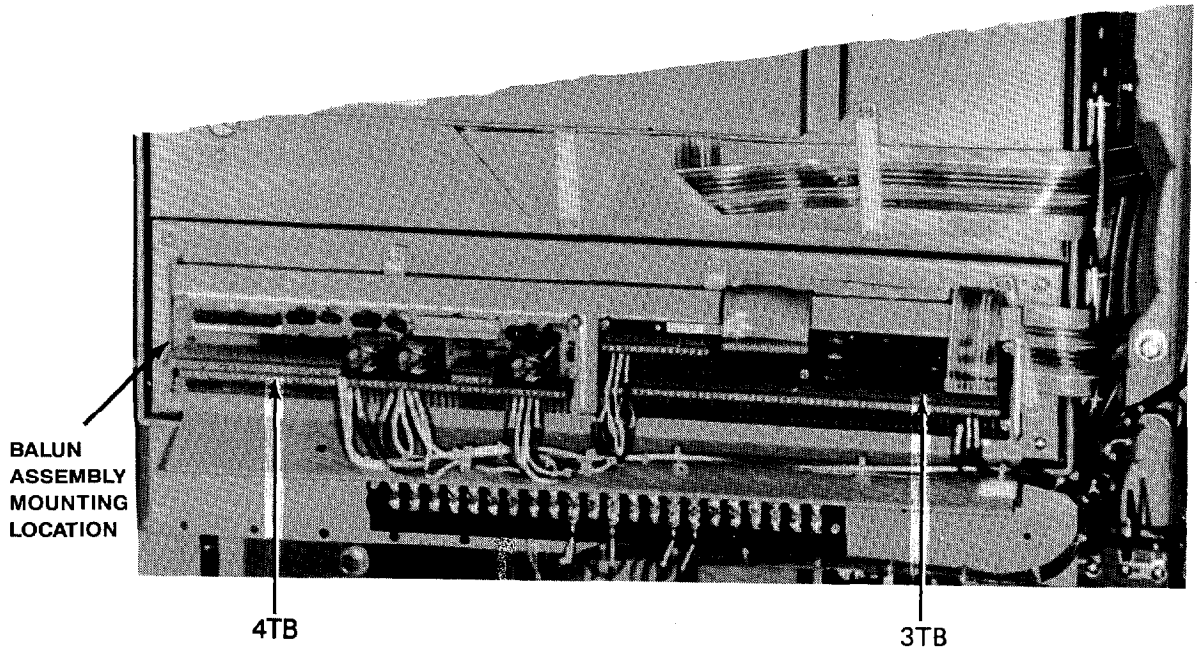


Figure 2. 4TB MOUNTING LOCATION ("D" Frame)

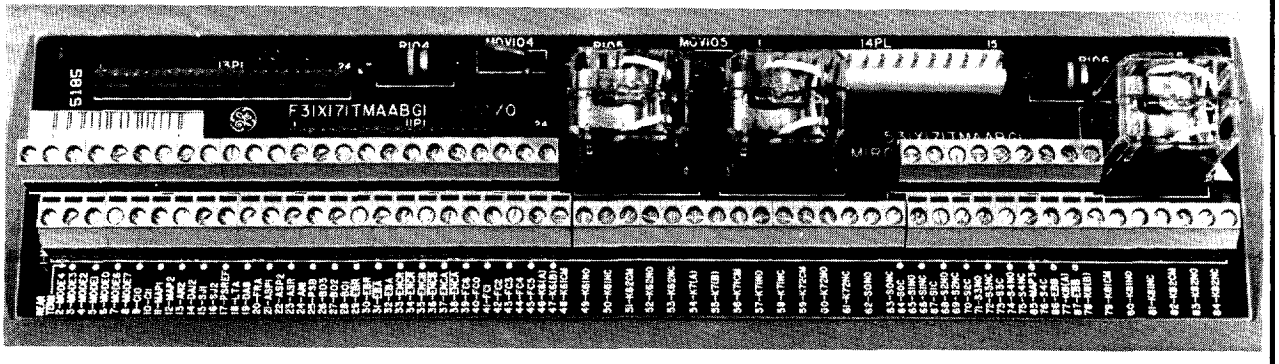


Figure 3. 4TB TERMINAL BOARD CARD

KIT INSTALLATION PROCEDURE

WARNING - SHOCK AND BURN HAZARD:
Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.

1. Verify that power to the controller that the option kit is to be installed is OFF.

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

2. Attach 4TB Terminal Board card to mounting screws (already installed on controller) as follows (see Figure 1 or Figure 2 for location):
   a. Place one mounting standoff over each of the four mounting screws.
   b. Place insulation barrier onto the four mounting screws.
   c. Remove 4TB Terminal Board card from anti-static bag and orient card with termination points towards bottom of mounting bracket and place onto the four mounting screws.
   d. Remove mounting hardware from shipping bag and slide a washer over each mounting screw and thread a nut onto each mounting screw.
   e. Tighten all mounting nuts.
3. Make desired connections to 4TB Terminal Board card (see Table 1 for connector pinout and terminal point descriptions).

TABLE 1: 4TB PINOUT AND TERMINAL POINT DESCRIPTIONS

CONNECTOR 11PL -- TO OPTIONAL APPLICATION CARD

Pin	4TB Term. #	Nomen-clature	Description
1	30	EBR	No connection
2	29	/EBR	No connection
3	85	MAP3	No connection
4	25	PSB	Strobe for optional 16 bit parallel input on 16PL. (See GEK-45167 Table 1, Application card, JP7.) Inputs must be set 50 microseconds before strobe, and strobe held for 100 microseconds.
5	26	B03	Optional programmable 24V bit output #3
6	27	B02	Optional programmable 24V bit output #2
7	28	B01	Optional programmable 24V bit output #1
8	32	EBA	No connection
9	31	/EBA	No connection
10	86	EBB	No connection
11	87	/EBB	No connection
12	33	ENCR	Not Currently Used
13	34	/ENCR	Not Currently Used
14	35	ENCB	Not Currently Used
15	36	/ENCB	Not Currently Used
16	37	ENCA	Not Currently Used
17	38	/ENCA	Not Currently Used
18	39	FC6	Bit 6 of optional fault code 24V output
19	40	FC0	Bit 0 of optional fault code 24V output
20	41	FC1	Bit 1 of optional fault code 24V output
21	42	FC2	Bit 2 of optional fault code 24V output
22	43	FC3	Bit 3 of optional fault code 24V output
23	44	FC4	Bit 4 of optional fault code 24V output
24	45	FC5	Bit 5 of optional fault code 24V output

TABLE 1: 4TB PINOUT AND TERMINAL POINT DESCRIPTIONS (cont.)

CONNECTOR 13PL -- TO OPTIONAL APPLICATION CARD

NOTE: *Refer to the Programming Software Package for definitions of I/O through 13PL for particular application.*

Pin	4TB Term. #	Nomen-clature	Description
1	6	MODE 0	Not Currently Used
2	5	MODE 1	Not Currently Used
3	4	MODE 2	Not Currently Used
4	3	MODE 3	Not Currently Used
5	2	MODE 4	Not Currently Used
6	1	MODE 5	Not Currently Used
7	7	MODE 6	Not Currently Used
8	8	MODE 7	Not Currently Used
9	9	CI0	No connection
10	10	CI1	No connection
11	11	MAP1	No connection
12	12	MAP2	No connection
13	13	AN2	No connection
14	14	DA12	Optional analog output from 12-bit D-A converter. Range = -10V to +10V, 1 MA, with 450 ohms output impedance. Update rate is 30 to 40 times/second.
15	15	SJ1	Not Currently Used
16	16	SJ2	Not Currently Used
17	17	PSREF	Optional analog position or process reference input ($\pm$10V).
18	18	LTA	Not Currently Used
19	19	DA8	Optional analog output from 8-bit D-A converter. Range = -10V to +10V, 1 MA, with 450 ohms output impedance. Normal update rate of 180 times/second; reduced to 30 to 40 times/sec if fast update selected for DA0.
20	20	FRA	Not Currently Used
21	21	ASP1	Not Currently Used
22	22	ASP2	Optional spare analog input #2 (programmable).
23	23	ASR	Optional auto speed reference input to drive, $\pm$15V maximum, positive for forward cell firing while motoring. ASR is the active speed reference in the auto mode when analog auto reference is selected.
24	24	AN1	No connection

TABLE 1: 4TB PINOUT AND TERMINAL POINT DESCRIPTIONS (cont.)

CONNECTOR 14PL -- TO OPTIONAL APPLICATION CARD

Pin	4TB Term. #	Nomen-clature	Description*
1	62	S0NO	Optional SLD #0 normally open contact.
2	63	S0NC	Optional SLD #0 normally closed contact.
3	64	S0C	Optional SLD #0 common contact.
4	65	S1NO	Optional SLD #1 normally open contact.
5	66	S1NC	Optional SLD #1 normally closed contact.
6	67	S1C	Optional SLD #1 common contact.
7	68	S2NO	Optional SLD #2 normally open contact.
8	69	S2NC	Optional SLD #2 normally closed contact.
9	70	S2C	Optional SLD #2 common contact.
10	71	S3NO	Optional SLD #3 normally open contact.
11	72	S3NC	Optional SLD #3 normally closed contact.
12	73	S3C	Optional SLD #3 common contact.
13	74	S4NO	Optional SLD #4 normally open contact.
14	75	S4NC	Optional SLD #4 normally closed contact.
15	76	S4C	Optional SLD #4 common contact.

***Contact ratings:** 1 Amp @115 VAC; 28 VDC, 30 microamps @ 30 millivolts*

115 VAC RELAY CONNECTION POINTS (K6, K7, K8)

4TB Term. #	Nomen-clature	Description
46	K6 (A)	K6 relay coil
47	K6 (B)	K6 relay coil
48	K61CM	K6 common contact (set 1)
49	K61NO	K6 normally open contact (set 1)
50	K61NC	K6 normally closed contact (set 1)
51	K62CM	K6 common contact (set 2)
52	K62NO	K6 normally open contact (set 2)
53	K62NC	K6 normally closed contact (set 2)
54	K7 (A)	K7 relay coil
55	K7 (B)	K7 relay coil
56	K71CM	K7 common contact (set 1)
57	K71NO	K7 normally open contact (set 1)
58	K71NC	K7 normally closed contact (set 1)
59	K72CM	K7 common contact (set 2)
60	K72NO	K7 normally open contact (set 2)
61	K72NC	K7 normally closed contact (set 2)
77	K8 (A)	K8 relay coil
78	K8 (B)	K8 relay coil
79	K81CM	K8 common contact (set 1)
80	K81NO	K8 normally open contact (set 1)
81	K81NC	K8 normally closed contact (set 1)
82	K82CM	K8 common contact (set 2)
83	K82NO	K8 normally open contact (set 2)
84	K82NC	K8 normally closed contact (set 2)

***Contact ratings:** 2.4 Amp @115 VAC; 28 VDC, 1 milliamp @ 0.1 volt*



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GEK-45161 (9/90)

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